

Quantum Mechanics III

Homework 7

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Topic

Quantum State Tomography

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Prove the first displayed relation used in Quantum State Tomography. State all assumptions and conventions.

$$p_k = \text{Tr}(\rho E_k)$$

2. Calculate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$\rho(T) = TT^\dagger / \text{Tr}(TT^\dagger)$$

3. Interpret the third relation in two limiting regimes and explain the physical meaning of the result.

$$\ell(\rho) = \sum_k n_k \ln \text{Tr}(\rho E_k)$$

Show enough intermediate work that another student can reproduce the calculation.